

Received 14 July 2025, accepted 3 August 2025, date of publication 12 August 2025,
date of current version 26 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3597908

RESEARCH ARTICLE

EEG-Driven Machine Learning for Stroke Detection in High-Risk Patients

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ABSTRACT Stroke remains a leading cause of disability and mortality worldwide, highlighting the need for effective tools for early detection and intervention. Recent research has explored the use of bio-signals generated by the human body as indicators of stroke occurrence. Among these, Electroencephalography (EEG) has shown particular promise. EEG-based stroke detection offers a non-invasive, cost-effective, accurate, and portable solution. This paper investigates the use of Machine Learning (ML) techniques with EEG data to detect strokes. Four experimental setups were designed to evaluate different feature engineering methods: using all features, selecting features via a Decision Tree (DT) with varying thresholds, and applying Principal Component Analysis (PCA) and Independent Component Analysis (ICA) for dimensionality reduction. Results indicate that the fourth setup—applying ICA with both AdaBoost and XGBoost—yielded the best performance, achieving an accuracy of 89%, precision of 86%, recall of 100%, F1-score of 92%, and a Matthews Correlation Coefficient (MCC) value of 0.76.

INDEX TERMS Bio-signals, EEG, machine learning, stroke detection.

I. INTRODUCTION

Strokes continue to be a primary reason for disability and death around the globe. They occur when a blood vessel in the brain bursts or something blocks the blood supply to part of the brain [1]. Moreover, strokes are typically categorized into two main types: ischemic stroke and hemorrhagic stroke, which have different patterns of incidence and prognosis [1]. Every year, over 12.2 million new strokes occur. In addition, above the age of 25, one in four individuals will experience a stroke in their lifetime, which necessitates the development of early detection and intervention tools [2]. People experiencing stroke might not immediately recognize the symptoms [3]. While every

minute goes without treatment, ischemic stroke patients lose 1.9 million neurons [4]. In addition, providing treatment to patients faster by 15 minutes has a significant positive effect on the lifetime quality of stroke patients [5]. Therefore, it is essential to rapidly identify the stroke signs and provide the appropriate diagnosis as soon as possible [6]. In the last few years, the interest in exploring the use of bio-signals as potential predictors of stroke occurrence has increased. Bio-signals, also referred to as physiological signals, indicate the measurable electrical or chemical activities produced by the human body. Moreover, these signals provide valuable information about the physiological processes and functioning of various organs and systems. In the context of stroke detection, bio-signals are used to identify specific patterns or changes that may indicate an increased risk of stroke [7].

The associate editor coordinating the review of this manuscript and approving it for publication was Paolo Crippa¹.

Traditionally, strokes are diagnosed by brain scans and physical examinations, such as Magnetic Resonance Imaging (MRI) and Computerized Tomography (CT) scans [8]. Despite these techniques being effective, they are time-consuming and cannot be used continuously since they may lead to an increase in cancer risk [9]. In contrast, bio-signals are used for stroke detection, including Electroencephalography (EEG), Photoplethysmography (PPG), Electrocardiography (ECG), and Electromyography (EMG). As a brief background, EEG measures the electrical activity of the brain by recording the voltage fluctuations generated by the firing of neurons [10]. Moreover, EEG signals play a vital role in detecting strokes within the initial hours of stroke onset, recognizing the presence of visual-spatial neglect (SN) in stroke survivors, and decoding the gait intention of chronic stroke patients [11], [13]. The use of EEG signals as a tool for stroke detection offers a persuasive motivation for research in this field as it is non-invasive, transportable, cost-effective, and accurate, making it facily accessible in various healthcare settings. When compared with CT scans, which use ionizing radiation and may not be as easily available in all healthcare facilities, EEG signals present a safer and more accessible alternative to early stroke detection and monitoring [14].

The main objective of this study is to develop Machine Learning (ML) models using EEG signals to precisely detect strokes in the initial hours as early as possible. Through the use of artificial intelligence, this research aims to enhance stroke detection accuracy, efficiency, and timeliness. As well as to help elderly patients and patients with disabilities since their stroke symptoms may be subtle [15]. The patients can wear wearable EEG devices, which will be linked to their doctors. Hence, any unusual sign will alert the doctor immediately [16]. Additionally, wearable EEG devices can be used easily in ambulances, which will help the paramedics make a quick decision about whether the patient should go to a hospital with a stroke department or any nearby medical center.

Previous studies ignored exploring many promising ML algorithms and feature engineering techniques. Therefore, the key contribution of this research is to conduct different experimental setups to explore the distinctions between each experimental setup to stroke detection and assess the impact of each on the performance and accuracy of stroke detection based on EEG signals.

ORGANIZATION OF THIS WORK:

The structure of this paper is presented in Figure 1 and outlined as follows. Section II presents the literature review of related works. The materials and methodology are proposed in Section III. The results of extensive ML experiments and discussion are illustrated in Section IV. The future work suggestions are presented in Section V. Finally, the conclusion and future works are indicated in Section VI.



FIGURE 1. The paper structure.

II. RELATED WORK

A. STROKE DETECTION

This section presents a review of previous studies related to stroke detection using bio-signals. In particular, the articles reviewed were classified on the basis of signals used and combinations thereof. In addition, as a concise overview of the employed signals, EEG measures electrical activity in the brain, detecting neural patterns and diagnosing disorders such as epileptic seizures [17]. ECG measures the electrical activity of the heart, which aids in the diagnosis of cardiac disease and arrhythmias [18]. EMG examines muscle electrical activity and assists in the identification of neuromuscular disorders [19]. PPG measures variations in blood volume using fingertip sensors to monitor heart rate and detect irregularities in blood flow [20].

Table 1 illustrates comprehensive details of the reviewed studies in this section, including the year of publication, dataset type, and size, implemented ML models, and obtained results.

1) EEG SIGNALS

A study by Laghari et al. [21] proposes a Cyber-Physical System (CPS) based on a Support Vector Machine (SVM) model to monitor the pre-stroke patient's status and detect stroke occurrence in real-time. Furthermore, in a paper by Wilkinson et al. [22], the authors proposed an inexpensive (Muse) mobile prehospital stroke diagnosis system to identify the presence of stroke and measure its severity using a machine-learning Random Forest (RF) model. Similarly, a study by Choi et al. [23] developed a health monitoring system that can anticipate the symptoms of stroke diseases for old people in real-time using the RF algorithm. Further, in a study conducted by Islam et al. [24], the authors present a predictive stroke model using EEG signals and a variety of ML algorithms.

2) ECG SIGNALS

In a study by Rathakrishnan et al. [25], the authors built a classification model to diagnose stroke disease using highly ranked ECG characteristics.

3) EMG SIGNALS

In a paper by Yu et al. [26], the authors proposed a system for detecting and predicting strokes using real-time EMG

TABLE 1. Summary of studies on ML methods for stroke prediction.

Ref	Year	Data Type	Dataset Size	ML Methods	Results
[21]	2018	EEG	30 Stroke Patients	SVM	Precision: 100%, Recall: 99.16%
[22]	2020	EEG	25 Stroke Patients	RF	Accuracy: 76%, Specificity: 86%, Sensitivity: 63%
[23]	2021	EEG	61 Stroke Patients and 212 Control Patients	RF	Accuracy: 92.51%
[24]	2022	EEG	48 Stroke Patients and 75 Control Patients	AdaBoost, XGBoost, and LightGBM	Accuracy (AdaBoost): 80%
[25]	2020	ECG	52 Stroke Patients and 80 Control Patients	KNN, RF, SVM, NB, and Logistic Regression	Accuracy (KNN): 96.6%, Accuracy (RF): 94.4%, Accuracy (SVM): 85.4%, Accuracy (NB): 72.7%, Accuracy (Logistic Regression): 66.9%
[26]	2020	EMG	287 Stroke Patients and 271 Control Patients	RF	Accuracy (RF): 90.38%
[27]	2024	EMG	48 Stroke Patients and 75 Control Patients	GBoost, RF, and HistGBoost	Accuracy (GBoost): 85.26%, Accuracy (RF): 85.30%, Accuracy (HistGBoost): 84.2%
[28]	2021	PPG and ECG	287 Stroke Patients and 287 Control Patients	RF, and DT	Accuracy (RF): 97.51%, Accuracy (DT): 91.56%

bio-signals and RF model. One recent paper by Hussain and Jany [27] developed an ML framework to classify myoelectric patterns of stroke patients and healthy individuals.

4) COMBINATION OF SIGNALS

In a study by Yu et al. [28], the authors suggested a new approach that forecasts stroke disease and gives a semantic interpretation based on PPG and ECG multi-modal bio-signals collected in real-time.

B. DISCUSSION AND KNOWLEDGE GAP

Comparison based on the utilized ML algorithms and feature engineering techniques was conducted and illustrated in Table 2, which highlights the significant gaps in previous studies that our paper aims to address. The prior studies were focused on limited ML algorithms such as SVM, RF, Light Gradient Boosting Machine (LightGBM), Adaptive Boosting (AdaBoost), and eXtreme Gradient Boosting (XGBoost). Meanwhile, they ignored other promising ML algorithms, such as Gaussian Naive Bayes (Gaussian NB), Bernoulli Naive Bayes (Bernoulli NB), K-Nearest Neighbors (KNN), and Decision Tree (DT). Furthermore, only the study [24] used the Independent Component Analysis (ICA), while

TABLE 2. Comparison of ML methods and feature engineering techniques for stroke prediction.

Ref	[21]	[22]	[23]	[24]	Our Paper
SVM	✓	×	×	×	×
Gaussian NB	×	×	×	×	✓
Bernoulli NB	×	×	×	×	✓
LightGBM	×	×	×	✓	✓
AdaBoost	×	×	×	✓	✓
XGBoost	×	×	×	✓	✓
KNN	×	×	×	×	✓
DT	×	×	×	×	✓
RF	×	✓	✓	×	✓
ICA technique	×	×	×	✓	✓
PCA technique	×	×	×	×	✓
DT technique	×	×	×	×	✓
EEG	✓	✓	✓	✓	✓

earlier studies have been overlooking other feature engineering techniques, such as Principal Component Analysis (PCA) and DT feature selection, which can significantly enhance model performance. Our paper aims to fill these gaps by exploring a wider range of ML algorithms and feature extraction techniques. Moreover, our paper will provide a comprehensive discussion showing the most effective approach for stroke detection based on the development of different experimental setups.

III. MATERIALS AND METHODS

This study developed binary classification models for classifying stroke disease based on EEG signals using the Python programming language through Google Colab. The study employed four experimental setups, as shown in the processing pipeline in Figure 2. The first experimental setup used all features, while the second experimental setup used the features selected by DT with three different thresholds. On the other hand, the third and fourth experimental setups compute PCA and ICA feature reduction methods on the dataset. Moreover, for all setups, the datasets were split into 80% training and 20% testing using both Train-Test Split using a stratified setting and GroupKFold methods to ensure robust evaluation. This is done by ensuring that each segment is either in the train split or the test split, i.e., not split across them, which prevents subject-level leakage. Similarly, all setups developed the same models, including eight ML models: Gaussian NB, Bernoulli NB, LightGBM, AdaBoost, XGBoost, KNN, DT, and RF. Bayesian Optimization was then employed to find the optimal hyperparameters. After that, performance metrics were used to evaluate the models, including accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient (MCC), in addition to the confusion matrix for the best model in each experiment.

A. RAW DATASET STATISTICAL ANALYSIS

This study employed the Stroke EIT dataset [29], which consists of 48 EEG records with an average duration of

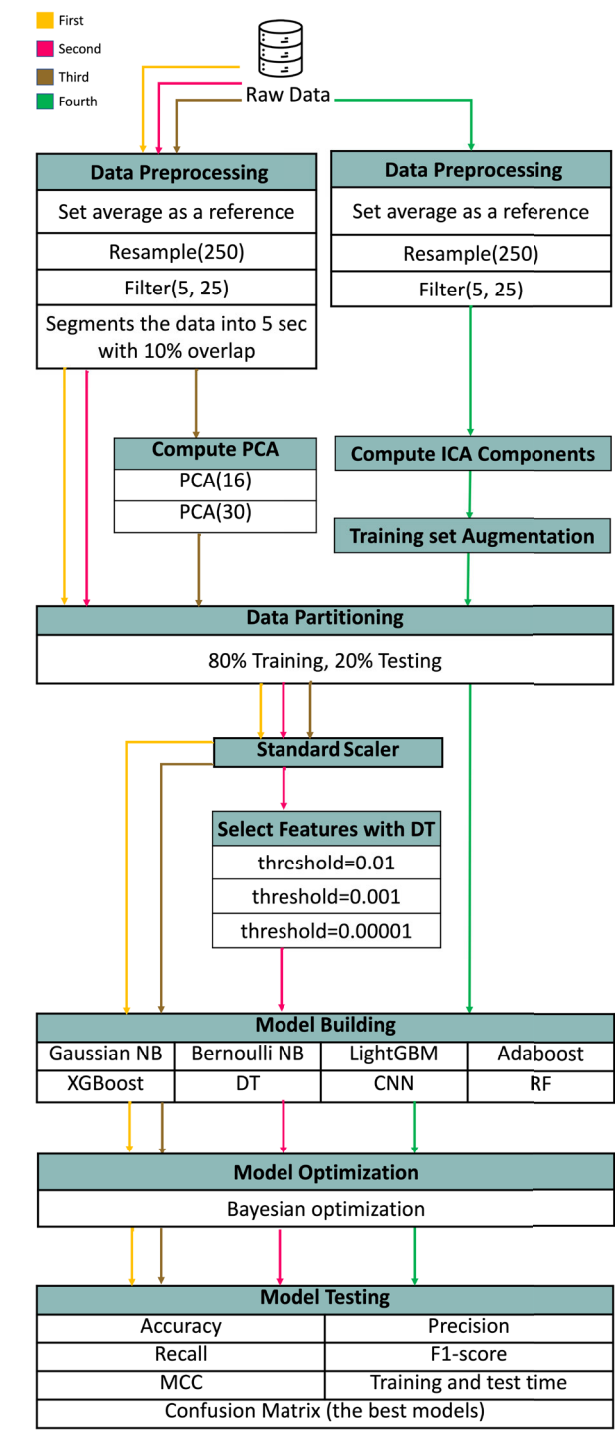


FIGURE 2. EEG signal processing pipeline.

12.17 minutes, including 20 records from healthy adult individuals and 28 from stroke adult patients, as shown in Figure 3. However, due to technical issues, three records have been excluded from the experiments. The dataset contains 34 features: 32 EEG channels, Time, and Stimulus, with the electrode positions illustrated in Figure 4. The EEG signals were recorded at a sampling frequency of 16384.0 Hz, using wet electrodes.

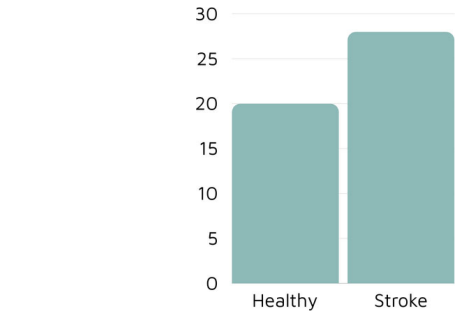


FIGURE 3. Class distribution.

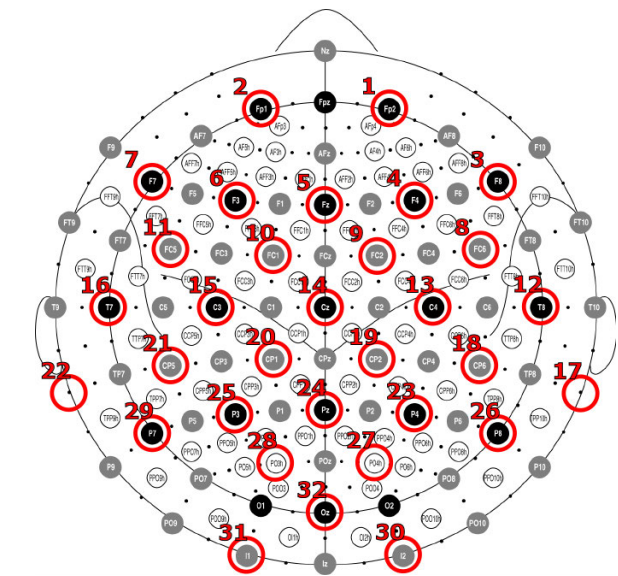


FIGURE 4. Electrodes (channels) position [29].

B. DATA PREPROCESSING

Data preprocessing is a fundamental step for developing AI models. The EEG dataset was preprocessed using the MNE package and the *scikit-learn* library. Below is a detailed description of the preprocessing steps:

- 1) **Setting the Reference** The raw EEG data was re-referenced to the average using the `set_eeg_reference()` function. This step helps minimize artifacts and is included in all experimental setups except for the fourth experimental setup, which employs ICA for artifact removal.
- 2) **Downsampling** The sampling rate of the raw data was initially 16,384 Hz, which was computationally intensive. To reduce complexity while retaining sufficient information for analysis, the data was downsampled to 250 Hz using the `resample(250)` function. This rate was chosen to ensure that the Nyquist criterion was satisfied, as the highest frequency in the observed signals was 124 Hz. Downsampling to 250 Hz provided a safe margin while optimizing computational efficiency.

- 3) **Filtering** The EEG data was filtered to a frequency range of 0.5–124 Hz using a bandpass filter implemented with the `firwin` method. The initial range of 1–45 Hz was selected based on prior studies and resources from a relevant GitHub repository. However, after assessing the results, it was found that this range did not yield satisfactory performance. Therefore, the range was extended to 0.5–124 Hz to better capture the critical EEG frequency bands while eliminating irrelevant noise. This adjustment improved the model’s performance by preserving the relevant signal features.
- 4) **Segmenting the Data** Continuous EEG signals were divided into overlapping epochs of 5 seconds with a 10% overlap. Segmentation into fixed-length epochs provides consistent input size for ML algorithms and enables temporal analysis. The 10% overlap was introduced to maximize data utilization and prevent information loss between consecutive epochs. A Hamming window was applied during segmentation to reduce edge artifacts caused by epoching.
- 5) **Dataset Split** The preprocessing resulted in a total of 45 samples before segmenting the data and 7,150 samples after segmentation for training and testing the ML algorithms. These samples were divided into 80% for training and 20% for testing.
- 6) **Standardization** Standardization was performed using the `StandardScaler()` function to normalize the features, ensuring that all input variables had a mean of 0 and a standard deviation of 1. This step was applied to all experimental setups except for the ML models in the fourth experimental setup (using ICA).

C. PREPROCESSED DATASET STATISTICS

EEG datasets are generally structured in a 3D format with the shape (samples/records, channels, time points). In this study, we worked with 45 EEG recordings, each comprising 33 channels. The number of time points for each recording was determined by the signal duration (in seconds) multiplied by the sampling frequency (in Hz). Before preprocessing, all 45 recordings contained 33 channels, while the number of time points varied due to differences in recording durations, averaging around 11,810,133 time points per recording. For the first, second, and third experiments, segmentation was applied, transforming the dataset from 45 recordings to 7,150 samples. Following downsampling, filtering, and segmentation, each sample was reduced to 1,250 time points. In contrast, the fourth experiment involved only downsampling and filtering without segmentation, resulting in samples that retained approximately 177,000 time points each after preprocessing.

To address the limited dataset size in the fourth experiment, data augmentation techniques were applied. These techniques included adding noise, scaling the amplitude, and shuffling the channels. The augmentation was performed separately for each class within the training set. As a result, the number of training samples increased from 36 to 216 after augmentation.

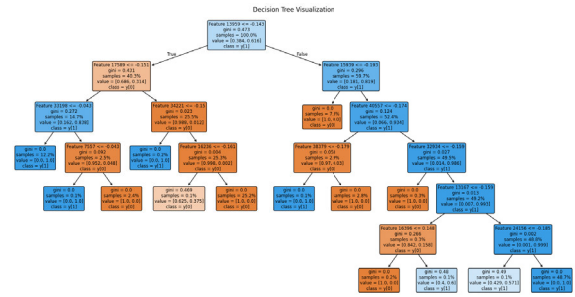


FIGURE 5. DT visualization for the second experimental setup.

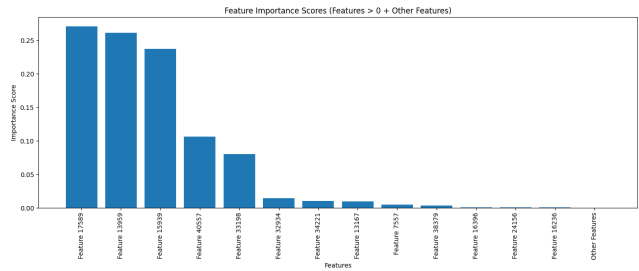


FIGURE 6. The relationship between the features and their importance scores.

D. FEATURE SELECTION TECHNIQUES

In the first experimental setup, all (33.1250) features are used. The other experiments applied the following techniques:

1) DESICION TREE (DT)

In the second experimental setup (using DT), only the selected features by DT are used. Since the DT model requires a fixed-length 1D feature vector for each sample, the 2D EEG feature matrix with shape (33, 1250) was flattened into a 1D vector of length 41,250 features. The feature selection procedure using the DT model involves three main steps. First, the DT classifier is trained, and its graphical representation is shown in Figure 5. Next, the most important features are selected based on the DT model using a threshold parameter, which determines the importance weights of the features. This study evaluates three threshold values: 0.01, 0.001, and 0.00001.

The number of selected features was determined based on their importance scores. As shown in Figure 6, the importance scores were plotted, revealing that only 10 features had an importance score greater than 0.

Our methodology involved incrementally refining the threshold by increasing its decimal precision and observing its effect on feature selection. Specifically, a threshold of 0.01 resulted in 6 features, and 0.001 yielded 9 features. Notably, while testing a threshold of 0.001, we consistently obtained 9 features. Therefore, we increased the threshold to 0.00001, which resulted in 10 features. Only features with an importance weight exceeding the threshold were retained as key features. Finally, the training and testing datasets

were transformed based on the selected features to ensure optimized model performance.

The selection of DT thresholds was determined through extensive experimentation, considering different numbers of selected parameters and their impact on model performance metrics such as accuracy and precision. The hyperparameters used in the feature selection process for building the DT models include max depth, min samples leaf, and criterion. In detail, the max depth was set to 20, the min_samples_leaf was set to 5, and the criterion for measuring the quality of a split was Gini impurity (criterion = 'gini'). These hyperparameters were constant throughout the experiment, while the DT threshold values were adjusted to determine the number of selected features.

2) PRINCIPAL COMPONENT ANALYSIS (PCA)

In the third experimental setup (using PCA), the features extracted by PCA are utilized. PCA is a mathematical technique for reducing the dimensionality of a dataset while preserving most of its important information. To reduce a dataset's dimensionality, PCA converts the original variables (features) into a new set of variables known as principal components, which are linear combinations of the original features. These primary components are orthogonal to one another and are ordered by the amount of variation they explain in the data [30]. In this experimental setup, 16 and 30 are used separately as the number of principal components retained from the original dataset.

The choice of retaining 16 and 30 principal components in this setup was based on an empirical evaluation of model performance. Multiple configurations were tested, and the optimal number of components was selected based on the results, considering metrics such as accuracy and computational efficiency. This manual selection process, although constrained by limited resources, ensured that the retained components preserved the dataset's essential information while reducing complexity.

3) INDEPENDENT COMPONENT ANALYSIS (ICA)

Lastly, in the fourth experimental setup (using ICA), the extracted features by ICA are used. ICA is a statistical method that decomposes multivariate signals into additive independent components. It is particularly effective for removing artifacts, extracting features, and reducing dimensionality in EEG data. EEG recordings often involve overlapping sensor readings, as multiple sensors on the scalp pick up signals from various sources. This is analogous to recording in a crowded area with multiple microphones capturing sounds from different sources. ICA separates these overlapping signals into statistically independent components, making it easier to isolate meaningful patterns [31].

For this study, the ICA implementation from the MNE Python library [32] was used. The FastICA

algorithm was employed, with the number of components (`n_components`) set to 32, based on the dimensionality of the dataset and the goal of preserving as much signal information as possible while reducing redundancy. A fixed random state (`random_state=42`) was used to ensure the reproducibility of the results.

The selection of 32 components was determined through manual experimentation and empirical testing. Different configurations were tested to evaluate their impact on classification performance. While resource constraints limited the ability to perform exhaustive hyperparameter tuning, this iterative approach allowed for fine-tuning of parameters and yielded satisfactory results.

While ICA is effective in separating overlapping sources and removing artifacts, it can also be sensitive to noise and may risk distorting meaningful EEG features. To address this, we applied band-pass filtering before running ICA, a standard practice that improves ICA stability by removing irrelevant low-frequency drifts and high-frequency noise that could otherwise mislead the decomposition.

E. DESCRIPTION OF ALGORITHMS

This study utilized several ML algorithms. The algorithms used include Gaussian NB, Bernoulli NB, LightGBM, AdaBoost, XGBoost, KNN, DT, and RF. These algorithms were selected due to their significant contributions to stroke prediction, as determined by a review of related studies. Notably, the ML algorithms mentioned were used in every experimental setup.

1) NAIVE BAYES (NB)

NB is a supervised ML algorithm based on Bayes' theorem. It calculates the probability of an instance belonging to a specific class and uses training data to learn the probabilities of its features. In the prediction step, the class with the highest probability is assigned. The "naive" assumption of feature independence allows NB to be fast and effective, especially in high-dimensional data scenarios. NB has three main types: Gaussian, Multinomial, and Bernoulli, each suited for different types of data. NB is commonly used for classification tasks, including signal data, due to its simplicity, speed, and ability to handle high-dimensional data with relative independence [33].

2) LIGHT GRADIENT BOOSTING MACHINE (LIGHTGBM)

LightGBM is a modern Gradient Boosted DTs algorithm that outperforms other boosting algorithms in terms of faster training, higher performance, lower memory usage, and improved prediction accuracy. It differs from traditional methods by employing a leaf-wise tree growth approach rather than a level-wise growth approach. In the leaf-wise method, nodes are expanded by adding leaf nodes, resulting in deep-narrow trees. This approach allows for faster tree expansion compared to the level-wise approach that creates shallow-wide trees [34].

3) ADAPTIVE BOOSTING (ADABOOST)

AdaBoost is a popular supervised ML algorithm that combines multiple weak learners to create a strong learner. It is effective for binary classification and can be used for both classification and regression problems. AdaBoost assigns higher weights to incorrectly classified instances, allowing subsequent weak learners to focus on those instances. This process is repeated until either the entire training dataset is accurately predicted or the maximum number of models is reached. While AdaBoost is fast and simple to implement, it can be sensitive to noise and outliers [35].

4) eXtreme GRADIENT BOOSTING (XGBoost)

XGBoost is a powerful ML algorithm that utilizes Gradient Boosted DTs. It excels in handling large datasets, managing missing values, and achieving high performance in regression and classification tasks. XGBoost's architecture relies on boosting, where models are sequentially built to correct the errors of previous models. The final prediction is obtained by combining the results through voting or averaging [36].

5) K-NEAREST NEIGHBORS (KNN)

The KNN method is a supervised ML algorithm used for classification and regression tasks. It predicts the class or value of a new data point by considering the majority class or average value of its k nearest neighbors in the training dataset. KNN selects the k closest training instances based on a distance metric, such as Euclidean distance, for classification and computes the average target value for regression. KNN is versatile, resistant to noise, and adaptable to changes in signal patterns. Its simplicity and ability to handle non-linear relationships make it useful for various applications in signal processing [37].

6) DECISION TREE (DT)

DT is another supervised ML algorithm that can be applied to both classification and regression tasks. It has a tree-like structure consisting of a root node, internal nodes, and leaf nodes. The algorithm begins at the root node and traverses down the tree, with each internal node representing an attribute used to split the data based on criteria such as information gain or Gini impurity. The process continues until a stopping criterion, such as reaching a maximum depth or having a minimum number of instances in a leaf node, is met. At that point, the algorithm assigns a predicted value or label to the leaf node. DTs are easy to implement and are robust in handling outliers without significant impact [38].

7) RANDOM FOREST (RF)

RF is a popular ensemble learning algorithm that utilizes multiple DTs to address supervised learning problems, such as regression and classification. It leverages the structure of DTs to make efficient predictions through majority voting or averaging. Both structures of the RF algorithm rely on DT, which consists of a root node that receives the training

data as input, and then, a series of questions at each node are created to support the best splitting of the training data until reaching the leaf node, which carries out the prediction. The algorithm's advantages include high accuracy with large and incomplete datasets, shorter training time, and avoidance of overfitting [39].

F. PERFORMANCE MEASURE

The performance of the developed models will be evaluated based on several performance evaluation measures, such as accuracy, precision, recall, F1-score, and MCC. Moreover, the training time and inference speed have been recorded. For each experiment, the confusion matrix of the best model is presented.

1) ACCURACY

Accuracy measures the percentage of correctly classified instances (stroke or non-stroke) out of the total number of instances. The formula is as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

2) PRECISION

Precision measures the proportion of correctly predicted stroke instances out of the total instances predicted as stroke. The formula is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

3) RECALL

Recall measures the proportion of correctly predicted stroke instances out of the total actual stroke instances. The formula is as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

4) F1-SCORE

The F1 score combines both precision and recall, providing a single metric that balances the trade-off between the two. It is a popular metric for imbalanced datasets because it assesses the model's performance in a way that accounts for the dataset's imbalanced nature, providing a more comprehensive evaluation of its ability to correctly identify stroke cases while minimizing mistakes. The formula is as follows:

$$F_1\text{-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

5) MATTHEWS CORRELATION COEFFICIENT (MCC)

MCC is a statistical measure used to evaluate the performance of binary classification models. It considers all four outcomes from the confusion matrix: true positives, true negatives, false positives, and false negatives. MCC is valid when the dataset is imbalanced, which is the case with the EIT dataset. MCC is considered a more balanced evaluation than accuracy. The MCC value ranges from -1 to $+1$. A value of $+1$ indicates perfect classification, while zero means no better than random

guessing, and -1 shows complete disagreement between predicted and actual classes. The formula is as follows:

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

6) TRAINING TIME AND INFERENCE SPEED

Training Time refers to the time an ML model needs to learn from the training data. It depends on several factors, including the dataset's size, the model's complexity, and the computational resources used. Faster training is often preferred in practical applications, especially when frequent model updates are needed. On the other hand, inference speed is the time it takes for a trained model to make predictions on new, unseen data. This metric is essential in real-time or resource-constrained environments, such as mobile applications or embedded systems. Models with high inference speed can respond more quickly, making them suitable for time-sensitive tasks. Both times have been measured in seconds.

7) CONFUSION MATRIX

A confusion matrix is a visualization of the model's predictions in a matrix. It shows true positives, true negatives, false positives, and false negatives. The matrix is as follows:

	Actual Positive	Actual Negative
Predicted Positive	TP	FP
Predicted Negative	FN	TN

G. OPTIMIZATION STRATEGY

To identify the optimal combination of hyperparameters, Bayesian optimization was conducted using the scikit-optimize package across eight models: Gaussian NB, Bernoulli NB, LightGBM, AdaBoost, XGBoost, KNN, DT, and RF. Bayesian Optimization is an efficient method for hyperparameter tuning in ML. Unlike grid search, which exhaustively evaluates all possible parameter combinations, Bayesian Optimization builds a probabilistic model of the objective function to guide the search process. It strategically balances exploration (evaluating new or uncertain areas of the parameter space) and exploitation (focusing on regions known to perform well), resulting in a more efficient and intelligent search compared to traditional methods [40].

Table 3 presents the hyperparameters and their respective search spaces used during Bayesian Optimization. Tables 4 and 5 present the optimal hyperparameter values obtained for the first (EX1), second (EX2), third (EX3), and fourth (EX4) experimental settings, respectively.

Tables 12 and 13 present the training and testing times (in seconds) required for the ML models.

IV. RESULTS AND DISCUSSION

A. COMPARATIVE EVALUATION OF THE FOUR EXPERIMENTS

This section presents and discusses the results of the ML models based on the four experimental setups.

TABLE 3. Hyperparameters and search spaces used in Bayesian optimization.

Model	Hyperparameter	Values
GaussianNB	var_smoothing	(1e-8, 1e-3)
Bernoulli NB	alpha binarize fit_prior	(0.1, 1.0) (0.2, 0.8) (True, False)
LightGBM	learning_rate n_estimators max_depth subsample colsample_bytree	(0.05, 0.2) (50, 100) (-1, 8) (0.6, 1.0) (0.6, 1.0)
AdaBoost	n_estimators learning_rate	(50, 100) (0.05, 0.2)
XGBoost	max_depth n_estimators learning_rate	(4, 8) (50, 100) (0.05, 0.2)
KNN	algorithm n_neighbors weights	(auto, ball_tree, kd_tree, brute) (1, 12) (uniform, distance)
DT	criterion max_depth min_samples_leaf	(gini, entropy) (3, 8) (2, 4)
RF	n_estimators max_features max_depth max_leaf_nodes	(50, 100) (sqrt, log2, None) (5, 10) (5, 10)

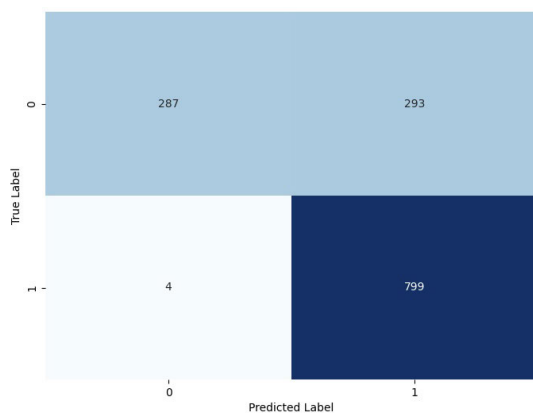
TABLE 4. Optimal hyperparameters for the first and second experimental setups.

Model	Hyperparameter	EX1 All	EX2 DT6	EX2 DT9	EX2 DT10
GaussianNB	var_smoothing	4.679e-05	1.1234e-06	1.1234e-06	1.1234e-06
Bernoulli NB	alpha binarize fit_prior	0.649 0.303 False	0.6877 0.7300 True	0.2785 0.7512 True	0.6492 0.3031 False
LightGBM	learning_rate n_estimators max_depth subsample colsample_bytree	0.155 63 6 0.831 0.601	0.0635 90 4 0.8092 0.9250	0.1701 98 2 0.9457 0.9350	0.1701 98 2 0.9457 0.9345
AdaBoost	n_estimators learning_rate	72 0.15	94 0.1596	94 0.1596	86 0.0883
XGBoost	max_depth n_estimators learning_rate	7 87 0.050	8 55 0.0926	6 76 0.1515	6 76 0.1515
KNN	algorithm n_neighbors weights	brute 3 distance	ball_tree 9 distance	brute 6 distance	brute 11 uniform
DT	criterion max_depth min_samples_leaf	entropy 4 3	gini 8 2	entropy 5 3	gini 7 4
RF	n_estimators max_features max_depth max_leaf_nodes	98 None 9 7	86 log2 9 8	90 sqrt 9 8	86 log2 9 8

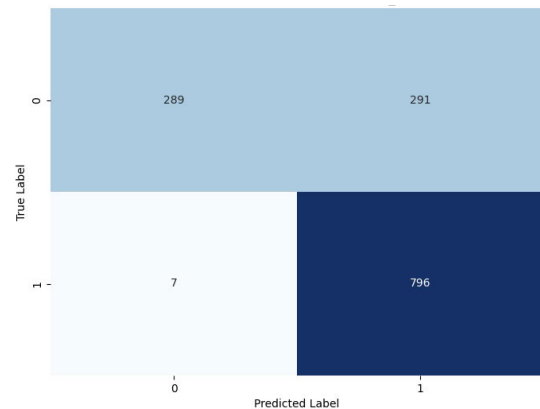
According to the findings of the first experiment in Table 6, whereas the models were trained on all available features without any feature selection methods, XGBoost and RF achieved the highest accuracy of 79% and identical MCC of 0.59. However, XGBoost outperformed others with a perfect recall of 100%, indicating that it detected almost all stroke cases (true positives) but had slightly lower precision, suggesting some false positives. From the confusion matrix of the XGBoost in Figure 7, XGBoost mistakenly classified four stroke samples as non-stroke (false negatives) and wrongly

TABLE 5. Optimal hyperparameters for the third and fourth experimental setups.

Model	Hyperparameter	EX3 PCA16	EX3 PCA30	EX4 ICA
GaussianNB	var_smoothing	9.9487e-05	9.949e-05	0.0002
Bernoulli NB	alpha	0.1008	0.542	0.6492
	binarize	0.6907	0.764	0.3031
	fit_prior	False	True	False
LightGBM	learning_rate	0.1839	0.179	0.1371
	n_estimators	59	92	66
	max_depth	0	3	7
	subsample	0.9202	0.730	0.8681
	colsample_bytree	0.8936	0.817	0.7640
AdaBoost	n_estimators	85	94	72
	learning_rate	0.1880	0.160	0.1515
XGBoost	max_depth	80	7	76
	n_estimators	5	94	6
	learning_rate	0.1542	0.188	0.1515
KNN	algorithm	brute	brute	brute
	n_neighbors	3	11	11
	weights	distance	uniform	uniform
DT	criterion	gini	entropy	entropy
	max_depth	8	5	7
	min_samples_leaf	2	3	3
RF	n_estimators	66	90	86
	max_features	None	sqrt	log2
	max_depth	7	9	9
	max_leaf_nodes	10	8	8

**FIGURE 7.** XGBoost confusion matrix from Experiment 1 using all features.

predicted 293 non-stroke samples as strokes (false positives). Furthermore, the RF confusion matrix in Figure 8 shows that the RF model mistakenly classified seven stroke samples as non-stroke (false negatives) and a smaller number of false positives (291 samples) than the XGBoost model. High recall is essential in medical applications where a false negative (missing a true stroke patient) is at least as bad as a false positive (over-reporting a non-stroke patient), consistent with the clinical goal of minimizing false negatives. However, with comparable recall rates, the precision was 73% for both XGBoost and RF, which implied that about a fourth of the positively predicted cases were negative. However, such false positives are often acceptable in medical applications to avoid the severe consequences of missed diagnoses. The MCC values of both XGBoost and RF were 0.59, the highest among

**FIGURE 8.** RF confusion matrix from Experiment 1 using all features.

all models in this experiment, indicating a strong balance between true and false predictions with the imbalanced dataset. In terms of computational efficiency, there was a significant difference in computational cost between XGBoost and RF at training time in terms of computational efficiency, as shown in Table 12. Specifically, RF took 8746.78 seconds for training. In comparison, XGBoost finished training in only 185.684 seconds, suggesting that the time required for RF to train the models was around 47 times higher than XGBoost, which implies a serious computational overhead. Such a difference is a result of the underlying architecture of the RF algorithm, which combines a large number of deep, independent DTs that are trained on a bootstrapped subset of the data. However, XGBoost implements a sequential boosting mechanism with regularization, resulting in reasonable tree growth and faster convergence. With this different training time in mind, we also note that testing (i.e., inference) times for both models were quite similar, with RF at 0.240 seconds and XGBoost slightly faster at 0.188 seconds. A slight difference in the speed of inference indicates that, although XGBoost has a clear advantage in terms of training efficiency, both models are acceptable for clinical decision support systems or other deployments via real-time or near-real-time testing in an environment where testing speed is important, such as stroke detection. The DT model yields results with an accuracy of 63%, precision of 74%, and recall of 55%. While its precision exceeded some of the better models, the relatively low recall indicates a higher number of false negatives, which is problematic in the clinical context of stroke detection. In addition, the recall of the LightGBM model is perfect, which is 100%, but the other metrics are low, including an accuracy of 58% and a precision of 58%. The imbalance between recall and precision significantly undermines its practical applicability despite capturing all stroke instances. Its MCC of 0.06 suggests that there is a random correlation between predictions and ground truth values. In terms of computational efficiency, the DT model requires less time for training than RF and is

relatively lower than XGBoost. As shown in Table 12, DT finished the training phase in 132.388 seconds, compared to 185.684 seconds for XGBoost and 8746.78 seconds for RF. This emphasizes the computational efficiency of DT, in which DT constructs only a single tree using the greedy splitting method without the need for the overhead of iterative optimization or ensemble methods. DT is obviously highly suitable for medical applications that require rapid model development or repeated training on growing and new data. However, while DT offers notable speed advantages during training, it typically does so at the cost of reduced accuracy and generalization, as it lacks the ensemble strength of XGBoost and RF. Therefore, in time-constrained environments where model accuracy is still critical—such as medical diagnosis—XGBoost may offer a better trade-off between efficiency and predictive power. The worst results in the first experiment are with AdaBoost, Gaussian NB, and KNN models. AdaBoost, although achieving a recall of 81%, suffered from low accuracy, 47%, and precision of 53%, which reflects its limited ability to distinguish between stroke and non-stroke signals. Its negative MCC, which is -0.27 , indicates that the predictions were more likely incorrect than correct, which is a red flag for any medical application. Gaussian NB yielded particularly poor results, with an accuracy of 47%, a recall of 46%, and an MCC of -0.06 . These metrics imply near-random classification, making it unsuitable for this experiment. The performance of the KNN model is insufficient, with an accuracy of 50%, a recall of 46%, and an F1-score of 52%. KNN's poor results may be attributed to the high dimensionality of EEG features, which degrades its performance due to the curse of dimensionality.

Moving on to the second experiment, where the DT method is used to select the most significant features, Table 7 presents the results for three different feature selection numbers: 6, 9, and 10. The AdaBoost model achieved the best performance in this experiment compared to the others, with 79% accuracy, 100% recall, 76% precision, and an F1-score of 87%, along with a 0.5 MCC value. AdaBoost yielded promising results, particularly in terms of recall, which is crucial for detecting stroke cases, with the highest MCC in this experiment, showing reliable classification. Furthermore, the AdaBoost confusion matrix in Figure 9 shows that the AdaBoost model mistakenly classified only two stroke cases as non-stroke (false negatives). However, 292 samples have been incorrectly identified as stroke cases.

However, none of the ML models improved when the number of selected features increased from 6 to 9. Moreover, when the number of selected features increased from 6 to 10, two models showed slightly better performance in all measured metrics: KNN and LightGBM. On the other hand, Bernoulli NB, AdaBoost, and XGBoost models exhibit a drop in performance, which could be due to several factors, such as overfitting, irrelevant features, or noisy features.

In terms of required training time, this experiment yielded the lowest training time for almost all trained

TABLE 6. First experiment results (using all features).

First Experimental Setup (Using All Features).					
Model	Accuracy	Precision	Recall	F1-score	MCC
Gaussian NB	47%	50%	46%	55%	-0.06
Bernoulli NB	54%	66%	44%	53%	0.13
LightGBM	58%	58%	100%	74%	0.06
AdaBoost	47%	53%	81%	64%	-0.27
XGBoost	79%	73%	100%	84%	0.59
KNN	50%	59%	46%	52%	0.01
DT	63%	74%	55%	63%	0.28
RF	79%	73%	99%	84%	0.59

TABLE 7. Second experiment results across different feature sets (using dt-selected features).

Model	Accuracy	Precision	Recall	F1-score	MCC
6 Features					
Gaussian NB	67%	69%	95%	80%	0.01
Bernoulli NB	68%	69%	99%	81%	0.00
LightGBM	68%	81%	69%	75%	0.33
AdaBoost	79%	76%	100%	87%	0.50
XGBoost	69%	82%	70%	76%	0.34
KNN	57%	69%	67%	68%	0.01
DT	69%	82%	70%	75%	0.34
RF	68%	81%	69%	75%	0.33
9 Features					
Gaussian NB	66%	68%	93%	79%	-0.02
Bernoulli NB	68%	69%	99%	81%	0.00
LightGBM	68%	81%	69%	75%	0.33
AdaBoost	78%	76%	99%	86%	0.48
XGBoost	68%	81%	69%	75%	0.33
KNN	57%	69%	67%	68%	0.02
DT	68%	81%	69%	75%	0.33
RF	68%	81%	69%	75%	0.33
10 Features					
Gaussian NB	67%	69%	96%	80%	0.01
Bernoulli NB	35%	64%	12%	20%	-0.04
LightGBM	79%	84%	85%	84%	0.50
AdaBoost	58%	69%	70%	70%	0.03
XGBoost	58%	70%	69%	70%	0.04
KNN	75%	75%	94%	84%	0.36
DT	68%	82%	69%	75%	0.34
RF	68%	81%	69%	75%	0.33

ML models when the number of features was 6, which indicates the importance of utilizing feature reduction methods.

The results of the third experiment are shown in Table 8, whereas the PCA is applied for reduced features (16 and 30 features). The KNN model performed relatively well with 16 features, achieving 77% accuracy, 94% precision, 71% recall, 81% F1-score, and 0.56 MCC.

The confusion matrix in Figure 10 shows that KNN correctly predicts 665 stroke cases and 393 non-stroke cases. On the other hand, 276 stroke cases are predicted as non-stroke (i.e., false negatives). However, the high precision of 94% implies that most of the cases were predicted correctly. The F1-score of 81%, which balances precision and recall, supports KNN's effectiveness in handling the class imbalance that is commonly observed in medical datasets.

The strong MCC of 0.56, which accounts for all elements in the confusion matrix, verifies the robustness of KNN's performance. Compared to other classifiers, this model maintains a good trade-off between sensitivity and specificity,

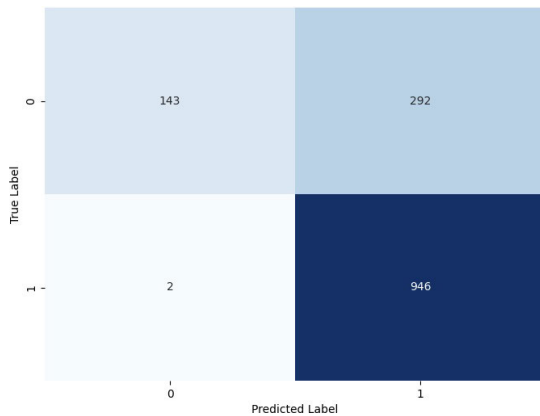


FIGURE 9. AdaBoost confusion matrix from Experiment 2 using six DT-selected features.

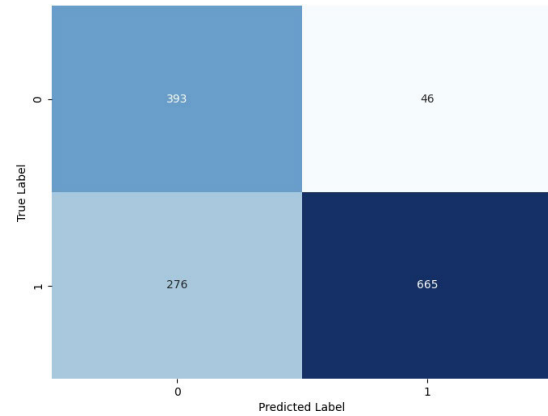


FIGURE 10. KNN confusion matrix from Experiment 3 using 16 PCA-selected features.

especially after PCA transformation to 16 features, which likely helped reduce noise and redundancy from the original high-dimensional EEG data. In addition, the AdaBoost model successfully predicts almost all stroke cases, as shown with 100% recall, which is preferred in medical applications, as failing to detect the stroke could lead to severe consequences. Despite AdaBoost's perfect recall, the results show low precision with only 68%, and the value of the MCC is 0.00, indicating that its predictions were not statistically correlated with the actual class labels. The high number of false positives significantly compromised the model's reliability. Although AdaBoost maximized the true positive rate, it misclassified a significant number of non-stroke samples as stroke cases. This could be due to the nature of the PCA transformation, which transforms the original feature space into a set of principal components. Even though PCA benefits reduce dimensionality and eliminate noise, it may also distort the localized decision boundaries that AdaBoost's weak learners, which typically rely on shallow DTs. Despite a relatively high F1-score of 81%, the inability of AdaBoost to yield balanced predictions makes it unsuitable for stroke detection systems as the records of non-stroke usually more than stroke cases.

When features are extended with PCA from 16 to 30, the performance of all models generally declines, indicating that adding more principal components may have reintroduced noise or irrelevant variance. For example, KNN's accuracy dropped to 57%, and its MCC fell sharply from 0.56 to 0.03, confirming its sensitivity to high-dimensional representations.

The fourth experimental setup employed ICA to reduce dimensionality and extract statistically independent components from the raw EEG signals. As we mentioned before, the data has not been segmented as in the previous experiments. The results of the fourth experiment setup are shown in Table 9. Among all models from the four experiments, XGBoost and AdaBoost achieved the highest classification performance, which was consistent across all metrics. Both

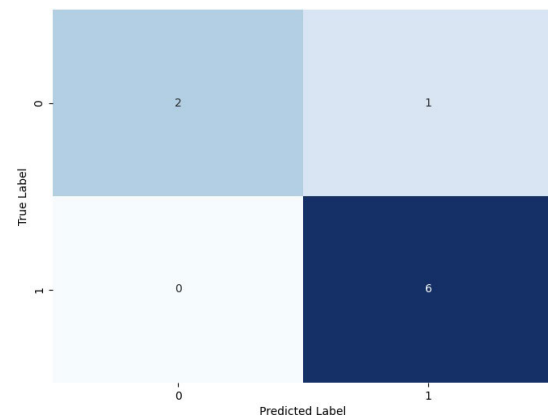


FIGURE 11. AdaBoost confusion matrix from Experiment 4.

models achieved an accuracy of 89%, a precision of 86%, a recall of 100%, and an F1-score of 92%, with an MCC of 0.76. The confusion matrices for AdaBoost and XGBoost in Figures 11 and 12 show that all six stroke cases were predicted correctly, and the non-stroke case was predicted as a stroke (i.e., false positive). Perfect recall, in which no stroke cases are missed, is a highly desirable result in the medical field, where false negatives are considered critical. On the other hand, the result of the precision is relatively acceptable, at 86%, which indicates a relatively low number of false positives. Furthermore, the MCC value of 0.76, which is the highest among all models in all four experiments, demonstrates a strong overall correlation between predicted and actual classes, demonstrating the robustness and reliability of both the XGBoost and AdaBoost models with the ICA. The effectiveness of these models can be attributed to their ability to capture complex non-linear interactions among the ICA components. XGBoost benefits from regularization and gradient boosting techniques, which enhance its generalization to noisy biomedical data. AdaBoost, by iteratively focusing on hard-to-classify samples, was also able to maintain high recall and F1 performance while minimizing bias.

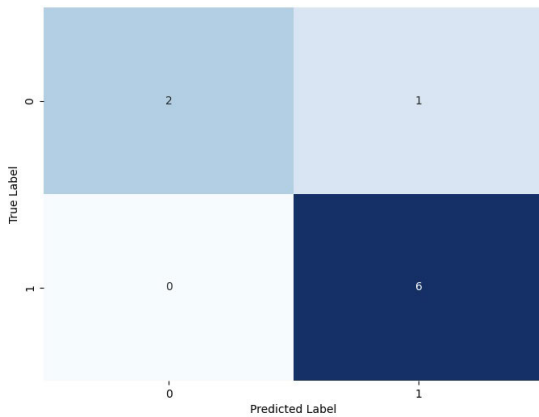


FIGURE 12. XGBoost confusion matrix from Experiment 4.

TABLE 8. Third experiment results (using PCA with 16 and 30 features).

Model	Accuracy	Precision	Recall	F1-score	MCC
16 Features					
Gaussian NB	58%	69%	69%	69%	0.03
Bernoulli NB	53%	99 %	31%	47%	0.34
LightGBM	57%	77%	52%	62%	0.17
AdaBoost	68%	68%	100%	81%	0.00
XGBoost	55%	76%	49%	60%	0.14
KNN	77%	94%	71%	81%	0.56
DT	65%	80%	65%	72%	0.29
RF	62%	75%	66%	70%	0.18
30 Features					
Gaussian NB	53%	62%	47%	54%	0.08
Bernoulli NB	52%	61%	44%	51%	0.06
LightGBM	49%	54%	80%	65%	-0.18
AdaBoost	51%	56%	73%	63%	-0.06
XGBoost	51%	55%	82%	66%	-0.14
KNN	57%	58%	93%	72%	0.03
DT	48%	54%	70%	61%	-0.13
RF	57%	58%	97%	72%	0.0069

TABLE 9. Fourth experiment results (using ICA).

Fourth experiment (using ICA)					
Model	Accuracy	Precision	Recall	F1-score	MCC
Gaussian NB	67%	80%	67%	73%	0.32
Bernoulli NB	67%	80%	67%	73%	0.32
LightGBM	78%	75%	100%	86%	0.50
AdaBoost	89%	86%	100%	92%	0.76
XGBoost	89%	86%	100%	92%	0.76
KNN	67%	67%	100%	80%	0.00
DT	44%	60%	50%	55%	-0.16
RF	78%	75%	100%	86%	0.50

B. COMPARATIVE DISCUSSION OF OPTIMAL MODEL PERFORMANCE PER EXPERIMENT

Figure 13 shows a performance comparison of the best models across all experiments based on accuracy, precision, recall, and F1-score measurements.

For all eight ML models tested with different feature extraction methods, the ML methods yielded better accuracy with ICA compared to PCA, DT-based feature selection, and non-feature selection (using all features), except for the DT model, which achieved better results with other feature selection methods. The MCC values of most ML models with ICA indicate a stronger balance between true

and false predictions with the imbalanced dataset than with other feature selection methods. This is because ICA has the capability of separating the source signal from the mixed signals in multi-channel time series data, such as EEG. In addition, EEG signals are mixtures of signals originating from different brain sources as well as artifacts (i.e., eye blinks, muscle activity, noise, and emotional response). Removing noise in EEG is essential for the ML model to learn from informative and clean EEG signals [41], [42], [43]. Concerning the results of the DT model with the ICA experiment, DTs are not suitable for learning from abstract statistical data, such as data extracted by ICA, which transforms the data into independent dimensions [44]. In terms of training time efficiency, it is reasonable to expect that reducing the number of selected features will result in a decrease in training time. This is observed in the results of DT-6 features, where seven models have the lowest training time compared to other experiments. Following that, the ML models with ICA rank second in terms of training time efficiency. It is notable that the AdaBoost and XGBoost models not only yield the best results with the ICA setting but also with other settings, such as AdaBoost with DT-6 features and XGBoost with All-Features. This is due to their robust learning methods, adaptive character, and built-in capability to prioritize valuable features while mitigating noise and redundancy. As is known, both AdaBoost and XGBoost are considered ensemble methods, both of which are based on DTs, giving them inherent advantages. In addition, XGBoost has the ability to address high-dimensional datasets by automatically selecting informative features via tree splits and regularizing complex branches to avoid overfitting to noisy or irrelevant features. These features enable XGBoost to exceed other models' performance with the All-Feature experiment. In addition to the fact that AdaBoost and XGBoost are able to capture non-linear relationships, both of them execute implicit feature selection.

In addition, the lower accuracy in Figure 13 is KNN with selected features based on PCA. This could be an indication that the number of selected features is insufficient. However, due to resource limitations, increasing the number of features was not feasible in this study.

Overall, based on the experimental setups conducted, the results in terms of the accuracy, recall, and training time of the applied feature engineering techniques in the second, third, and fourth experimental setups demonstrate an improvement in the overall performance of the ML model compared to the first experimental setup, which involved building ML models with all features. Therefore, the results emphasize the effectiveness of DT feature selection, PCA, and ICA techniques in improving the performance of ML models by training them with the most significant features extracted from EEG signals, thus reducing the noise and dimensionality of EEG signals. In addition, DT, PCA, and ICA techniques require less training and testing time compared to using all features, which is to be expected given the reduced dimensionality of the input data.

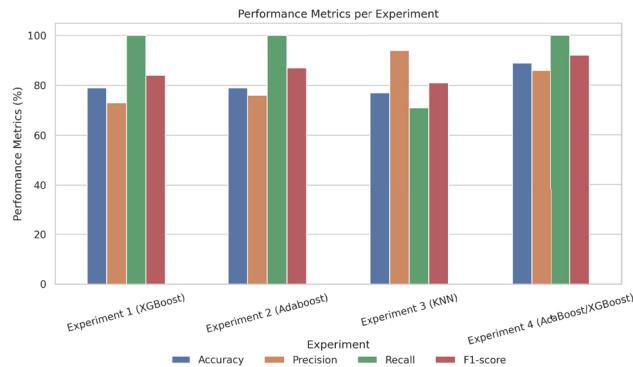


FIGURE 13. Comparison of the best models.

ICA is considered an effective EEG data analysis technique due to its ability to remove EEG data artifacts and identify the most informative features [45], [46]. This makes it more suitable than PCA, which focuses on variance rather than independence, making it less effective in separating meaningful signals. However, ICA is known to be sensitive to noise, which can affect its performance. To mitigate this limitation in our study, we applied filtering to reduce noise in the EEG data before applying ICA. This step ensured the robustness of the ICA feature reduction process and improved its effectiveness in identifying meaningful features. Similarly, AdaBoost and XGBoost are well-suited for small datasets like ours, as they combine multiple weak classifiers into a single robust predictive model. This ensemble approach allows AdaBoost and XGBoost to outperform other implemented ML models, achieving superior accuracy. These results demonstrate the superior efficiency of the ICA technique in preprocessing EEG data, as well as the AdaBoost and XGBoost models in classifying EEG signals for stroke detection, making them the best-performing models over other implemented techniques in this study.

C. EVALUATION AND EXPLAINABILITY OF TOP PERFORMING MODELS

Most ML models are considered black-box models because their predictions provide little to no explanation of how decisions are made. However, understanding the model's decision-making process is essential in diagnosis applications such as stroke detection to ensure trust and reliability. The models that produce the best performance in this study are AdaBoost and XGBoost, which utilize the ICA feature selection method.

To improve interpretability, we implemented SHapley Additive exPlanations (SHAP) values to explain the contribution of each feature to the model's prediction. Figures 14 and 15 illustrate the top features selected by the ICA method, shown on the left y-axis and ordered by their mean absolute SHAP value, regardless of direction (positive or negative), which means the features are ranked from the top based on the importance in predicting Class 1 (Stroke).

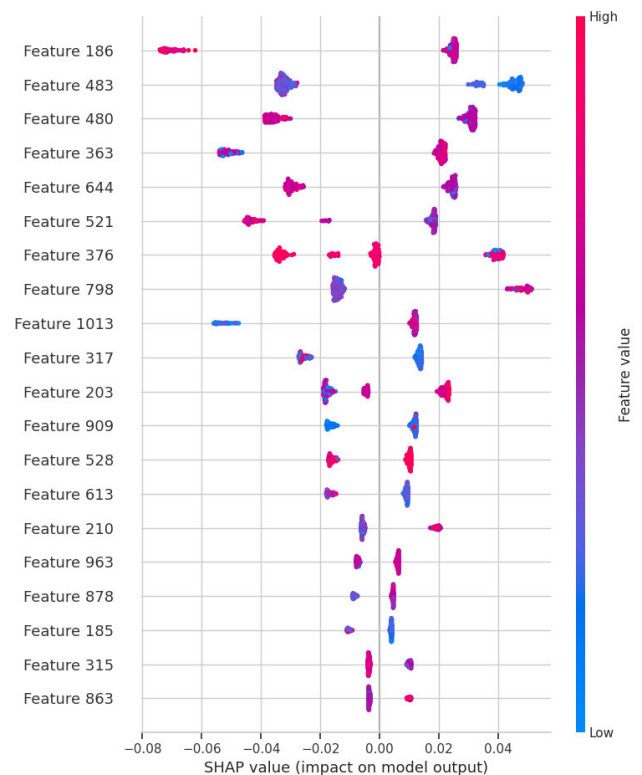


FIGURE 14. Class 1 SHAP values for the best model (AdaBoost with ICA).

The right y-axis represents the actual feature values, where red indicates high values and blue indicates low values. The x-axis displays the SHAP values, which reflect each feature's impact on the model's prediction. Positive SHAP values indicate a stronger contribution toward predicting Stroke (Class 1), while negative SHAP values reflect a contribution toward predicting Healthy (Class 0). As the ICA transforms a multivariate dataset into independent components, it is challenging to determine the interpretation of these features.

However, it is important to mention that the mean absolute SHAP value does not necessarily cause the prediction of stroke. It means the feature has a large influence on the decision, whether it increases or decreases the stroke probability. For example, although Feature 186 has the highest absolute SHAP value of the AdaBoost model in Figure 14, the model is strongly influenced by Feature 186 when predicting non-stroke cases, as high feature values (i.e., red dots) are on the left side (i.e., negative SHAP values), which indicates Feature 186 reduces stroke detection. On the other hand, Feature 483 has a strong impact on detecting stroke cases because it has a high positive SHAP value when the feature value is high.

One of the popular methods for evaluating the binary classification model is the AUC-ROC Curve. The ROC (Receiver Operating Characteristic Curve) is a graph that plots the true positive rate, which is known as Recall, against the false positive rate at different classification threshold

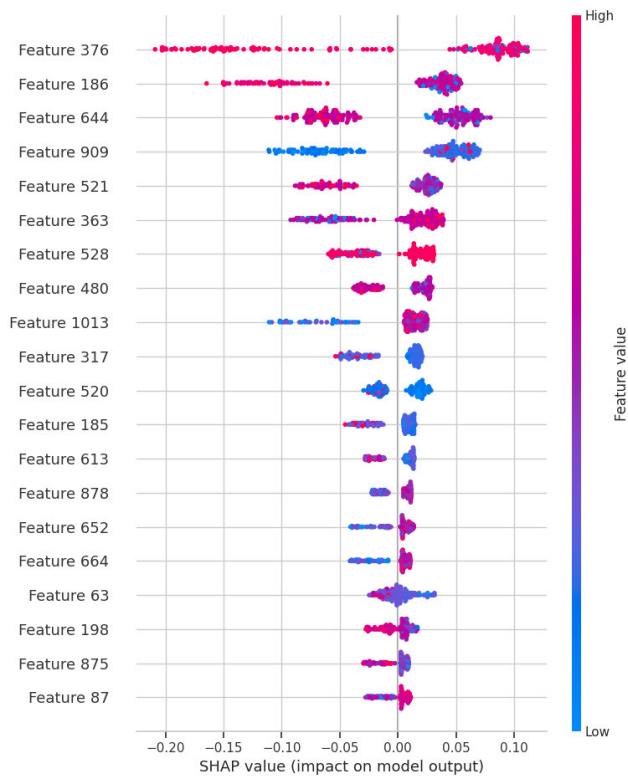


FIGURE 15. Class 1 SHAP values for the best model (XGBoost with ICA).

levels. It visualizes the model's ability to distinguish between two classes: stroke and non-stroke, in our case. On the other hand, AUC (Area Under the Curve) measures the area under the ROC curve. A higher AUC value means a more suitable model performance as it presents a greater ability to distinguish between classes.

As shown in Figure 16, the ROC curve of the AdaBoost model with ICA shows an AUC value of 0.83. In addition, the Figure 17 shows the ROC curve of the XGBoost model with ICA, which shows an AUC value of 0.94. Both AUC-ROC curves indicate robust overall discriminative power, especially for XGBoost. Both curves increase sharply to the top-left, which means the model maintains high sensitivity even at low false positive rates. This is important in clinical fields such as stroke detection, where maximizing the true positive rate while minimizing missed cases is critical.

As AdaBoost and XGBoost with ICA were the best-performing models in our study, we further enhanced the evaluation by incorporating 95% confidence intervals (CI) to assess the model's reliability over the long term. As shown in Table 10, the AdaBoost model achieved 89% accuracy (95% CI: 0.44 – 1), 86% precision (95% CI: 0.5 – 1), 100% recall (95% CI: 0.5 – 1), and 92% F1-score (95% CI: 0.5 – 1). On the other hand, the XGBoost model achieved 89% accuracy (95% CI: 0.66 – 1), 86% precision (95% CI: 0.57 – 1), 100% recall (95% CI: 1 – 1), and 92% F1-score (95% CI: 0.73 – 1). Even though both AdaBoost

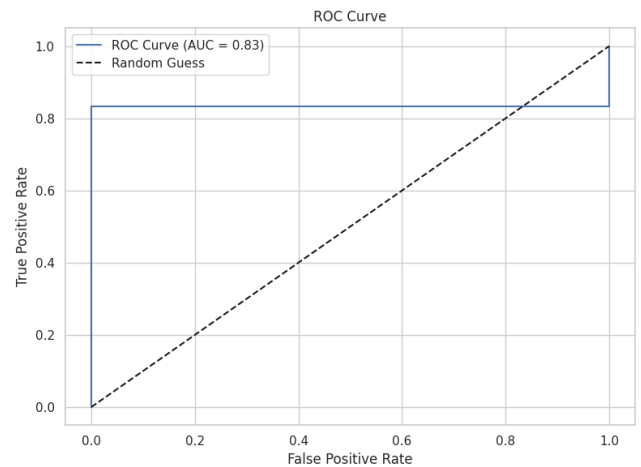


FIGURE 16. AUC-ROC curve for the best model (AdaBoost with ICA).

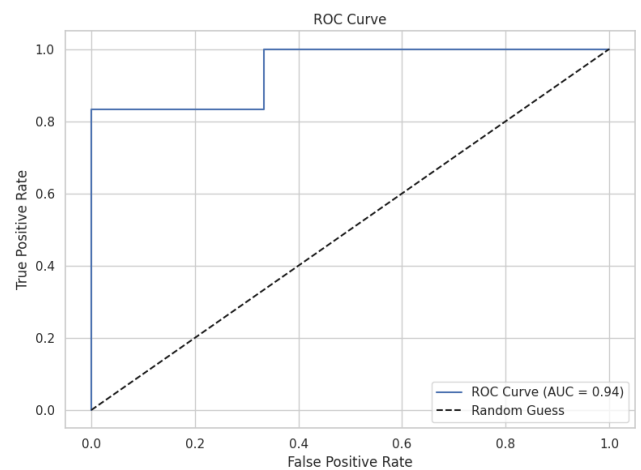


FIGURE 17. AUC-ROC curve for the best model (XGBoost with ICA).

and XGBoost models result in the same accuracy, precision, recall, and F1-Score, XGBoost is more stable and dependable based on CI results, as shown in Table 10. These CIs of the XGBoost model confirm the stability of the model's performance, especially in recall, which is vital in stroke prediction.

The variability in AdaBoost may arise from its sensitivity to noisy and outlier data, a known issue with AdaBoost. On the other hand, XGBoost's regularization, tree pruning, and gradient boosting enhancements help it to be more robust to variance.

D. REAL-WORLD APPLICATION

The results of our study demonstrate promising performance in detecting stroke, emphasizing the possibility of incorporating the ML model into portable diagnostic techniques for early stroke detection. Compared to traditional methods, such as CT scans and MRIs, portable diagnostic devices have benefits in terms of cost-effectiveness and rapid diagnosis.

TABLE 10. Performance metrics with 95% CIs for AdaBoost and XGBoost with ICA.

Metric	Value	95% CI Lower	95% CI Upper
AdaBoost Model			
Accuracy	0.89	0.44	1.0
Precision	0.86	0.5	1.0
Recall	1.0	0.5	1.0
F1-Score	0.92	0.5	1.0
XGBoost Model			
Accuracy	0.89	0.66	1.0
Precision	0.86	0.57	1.0
Recall	1.0	1.0	1.0
F1-Score	0.92	0.73	1.0

For example, as stated in the previous study [22], the device requires only five minutes to set up, making it possible for use in ambulances. However, a critical limitation of this approach is the need for a large and diverse dataset to handle signal variability across individuals and ensure accurate classification. Furthermore, the system should be user-friendly, minimizing reliance on specialized operators and thereby addressing challenges related to setup and interpretation by non-expert users. The integration of wearable EEG headset devices into ambulances offers a compelling real-world application with significant possibilities to enhance emergency care by enabling paramedics to make timely, data-driven decisions. For successful real-world implementation, several practical challenges should be considered carefully. The first is that real-time data processing remains a critical hindrance, as EEG signal analysis often requires high computational resources that may be difficult to support in compact, wearable devices. Additionally, EEG signals are highly susceptible to noise and artifacts, especially in dynamic environments like ambulances, where patient movement, vibrations, and environmental interference can degrade signal quality. Moreover, wearable devices encounter limitations in terms of battery life, processing power, and data storage capacity, all of which can hinder continuous monitoring and real-time decision-making. Addressing these technical and operational constraints is essential to ensure the proposed solution's feasibility, reliability, and clinical utility.

E. LIMITATIONS

Although our study achieved remarkable results, some limitations need to be addressed in future work. First, our study was performed on a small dataset, which may affect the generalizability of the results, emphasizing the need for larger, more diverse datasets to enhance model robustness. Second, the proposed approach has been conducted with low computational resource efficiency, which may delay the performance of real-time portable diagnostic systems for early stroke prediction.

TABLE 11. Acronyms and their definitions.

Acronym	Definition
AI	Artificial Intelligence
Bio-signals	Biological Signals
EEG	Electroencephalography
ECG	Electrocardiography
PPG	Photoplethysmography
EMG	Electromyography
MRI	Magnetic Resonance Imaging
CT	Computerized Tomography
ML	Machine Learning
CPS	Cyber Physical System
SVM	Support Vector Machine
RF	Random Forest
DT	Decision Tree
KNN	K-Nearest Neighbors
XGBoost	eXtreme Gradient Boosting
AdaBoost	Adaptive Boosting
LightGBM	Light Gradient Boosting Machine
NB	Naive Bayes
MCC	Matthews Correlation Coefficient
PCA	Principal Component Analysis
ICA	Independent Component Analysis
SHAP	SHapley Additive exPlanations
CI	Confidence Interval
AUC-ROC	Area Under the Curve - Receiver Operating Characteristic

V. FUTURE WORK

In the future, it would be beneficial to expand the size of the datasets used in this study to include more records from larger-sized individuals, both healthy and stroke-diagnosed, to further increase the robustness and generalizability of the models. In the study [25], the results achieved high accuracy using ECG features for post-stroke classification, which highlighted another area of future work is to make use of ECG instead of EEG to examine the potential impact it can have on stroke detection. We acknowledge that EEG offers a more direct method for detecting stroke as it measures brain activity. However, ECG signals could provide a significant benefit in real healthcare applications due to their high availability and ease of measurement. For example, wearable devices, such as smartwatches, have the capability to capture ECG signals seamlessly, which makes ECG a possible choice for incorporation into wearable devices. Additionally, combining EEG and ECG signals could improve the detection of the medial case by providing complementary insights, which leads to improved model robustness.

VI. CONCLUSION

As cases of stroke diagnoses increase worldwide, it is crucial to invent tools to accurately and safely detect strokes. The presented study aimed to utilize EEG signals to identify patterns indicating an increased stroke risk. Eight ML models were used in this paper. Specifically, the Gaussian NB, Bernoulli NB, LightGBM, AdaBoost, XGBoost, KNN, DT, and RF. To develop the previous models, the Stroke EIT dataset, comprising 45 EEG records from 33 healthy and stroke individuals, was utilized. Bayesian Optimization was applied to find the optimal parameters to increase the

TABLE 12. Training time and testing speed for the first and second experiments (in Seconds).

Model	Phase	EX1 All	EX2 DT6	EX2 DT9	EX2 DT10
GaussianNB	Training	3.230	0.115	0.136	0.117
	Testing	1.104	0.001	0.002	0.002
Bernoulli NB	Training	2.745	0.118	0.132	0.140
	Testing	0.992	0.001	0.002	0.002
LightGBM	Training	99.16	0.152	0.152	0.149
	Testing	0.0.215	0.005	0.006	0.005
AdaBoost	Training	2790.436	0.320	0.363	0.425
	Testing	11.296	0.032	0.039	0.032
XGBoost	Training	185.684	0.147	0.166	0.151
	Testing	0.188	0.003	0.003	0.004
KNN	Training	5.508	0.268	0.315	0.196
	Testing	12.166	0.103	0.030	0.074
DT	Training	132.388	0.120	0.132	0.138
	Testing	0.210	0.001	0.001	0.001
RF	Training	8746.78	0.262	0.315	0.359
	Testing	0.240	0.0.012	0.0.015	0.0.021

TABLE 13. Training time and testing speed for the third and fourth experiments (in Seconds).

Model	Phase	EX3 PCA16	EX3 PCA30	EX4 ICA
GaussianNB	Training	2.243	2.764	0.188
	Testing	0.0.513	0.985	0.002
Bernoulli NB	Training	1.674	2.368	0.184
	Testing	0.328	0.893	0.003
LightGBM	Training	70.292	52.873	1.1098
	Testing	0.055	0.084	0.004
AdaBoost	Training	1404.984	3292.926	3.261
	Testing	6.331	17.281	0.041
XGBoost	Training	78.657	331.2	3.935
	Testing	0.098	0.136	0.006
KNN	Training	5.369	17.031	0.225
	Testing	5.957	31.955	0.007
DT	Training	100.506	165.873	0.422
	Testing	0.106	0.154	0.001
RF	Training	3158.234	37.731	0.445
	Testing	0.129	0.225	0.020

performance and generalizability of the models. Furthermore, this study applied three feature engineering techniques, DT, ICA, and PCA, to extract meaningful information from the EEG records and reduce the dimensionality of the data for analysis. Several experiments were performed, including an experimental setup using all features, an experimental setup using DT, an experimental setup using PCA, and an experimental setup using ICA. The results indicated that AdaBoost and XGBoost with ICA yielded the best results, scoring an accuracy of 89%, a precision of 86%, a recall of 100%, an F1 score of 92%, and an MCC value of 0.76. The proposed model demonstrates strong potential for early stroke prediction using wearable EEG devices, providing a cost-effective and rapid alternative to traditional methods, such as CT and MRI, particularly in emergency settings, including ambulances. However, challenges such as the need for large, diverse training data, real-time processing limitations, signal noise, and hardware constraints must be addressed to ensure the reliable and practical deployment of these systems.

VII. ACRONYMS

See Table 11.

APPENDIX

TRAINING AND TESTING TIME

See Tables 12 and 13.

REFERENCES

- [1] A. Bersano and L. Gatti, "Pathophysiology and treatment of stroke: Present status and future perspectives," *Int. J. Mol. Sci.*, vol. 24, no. 19, 2023. [Online]. Available: <https://www.mdpi.com/1422-0067/24/19/14848>
- [2] V. L. Feigin, M. Brainin, B. Norrving, S. Martins, R. L. Sacco, W. Hacke, M. Fisher, J. Pandian, and P. Lindsay, "World stroke organization (WSO): Global stroke fact sheet 2022," *Int. J. Stroke*, vol. 17, no. 1, pp. 18–29, Jan. 2022.
- [3] K. W. Faiz, A. Sundseth, B. Thommessen, and O. M. Rønning, "Factors related to decision delay in acute stroke," *J. Stroke Cerebrovascular Diseases*, vol. 23, no. 3, pp. 534–539, Mar. 2014. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1052305713001742>
- [4] J. L. Saver, "Time is brain—Quantified," *Stroke*, vol. 37, no. 1, pp. 263–266, Jan. 2006.
- [5] A. Meretoja, M. Keshkaran, J. L. Saver, T. Tatlisumak, M. Parsons, M. Kaste, S. M. Davis, G. A. Donnan, and L. Churilov, "Stroke thrombolysis," *Stroke*, vol. 45, no. 4, pp. 1053–1058, Apr. 2014.
- [6] R. M. Zweifler, "Initial assessment and triage of the stroke patient," *Prog. Cardiovascular Diseases*, vol. 59, no. 6, pp. 527–533, May 2017.
- [7] M. Katan and A. Luft, "Global burden of stroke," *Seminars Neurol.*, vol. 38, no. 2, pp. 208–211, Apr. 2018.
- [8] P. C. Sanelli, J. B. Sykes, A. L. Ford, J.-M. Lee, K. D. Vo, and D. K. Hallam, "Imaging and treatment of patients with acute stroke: An evidence-based review," *Amer. J. Neuroradiology*, vol. 35, no. 6, pp. 1045–1051, Jun. 2014.
- [9] D. L. Lam, D. B. Larson, J. D. Eisenberg, H. P. Forman, and C. I. Lee, "Communicating potential radiation-induced cancer risks from medical imaging directly to patients," *Amer. J. Roentgenology*, vol. 205, no. 5, pp. 962–970, Nov. 2015.
- [10] D. Yoon, J.-H. Jang, B. J. Choi, T. Y. Kim, and C. H. Han, "Discovering hidden information in biosignals from patients using artificial intelligence," *Korean J. Anesthesiology*, vol. 73, no. 4, pp. 275–284, Aug. 2020.
- [11] L. Shreve, A. Kaur, C. Vo, J. Wu, J. M. Cassidy, A. Nguyen, R. J. Zhou, T. B. Tran, D. Yang, A. I. Medizade, B. Chakravarthy, W. Hoonpongmanont, E. D. Barton, W. Yu, R. Srinivasan, and S. C. Cramer, "Electroencephalography measures are useful for identifying large acute ischemic stroke in the emergency department," *J. Stroke Cerebrovascular Diseases*, vol. 28, no. 8, pp. 2280–2286, Aug. 2019.
- [12] J. Mak, D. Kocanaogullari, X. Huang, J. Kersey, M. Shih, E. S. Grattan, E. R. Skidmore, G. F. Wittenberg, S. Ostadabbas, and M. Akcakaya, "Detection of stroke-induced visual neglect and target response prediction using augmented reality and electroencephalography," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1840–1850, 2022.
- [13] J. Choi, H. Kang, S. H. Chung, Y. Kim, U. H. Lee, J. M. Lee, S. Kim, M. H. Chun, and H. Kim, "Detecting voluntary gait intention of chronic stroke patients towards top-down gait rehabilitation using EEG," *PubMed*, vol. 2014, pp. 1560–1563, Aug. 2016.
- [14] J. I. Doerrfuss, T. Kilic, M. Ahmadi, M. Holtkamp, and J. E. Weber, "Quantitative and qualitative EEG as a prediction tool for outcome and complications in acute stroke patients," *Clin. EEG Neurosci.*, vol. 51, no. 2, pp. 121–129, Mar. 2020.
- [15] Y. Athavale and S. Krishnan, "Biosignal monitoring using wearables: Observations and opportunities," *Biomed. Signal Process. Control*, vol. 38, pp. 22–33, Sep. 2017. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1746809417300617>
- [16] J. Andersson, Å. Rejnö, S. Jakobsson, P.-O. Hansson, S. J. Nielsen, and L. Björck, "Symptoms at stroke onset as described by patients: A qualitative study," *BMC Neurol.*, vol. 24, no. 1, p. 150, May 2024.
- [17] M. Soufneyestani, D. Dowling, and A. Khan, "Electroencephalography (EEG) technology applications and available devices," *Appl. Sci.*, vol. 10, no. 21, p. 7453, Oct. 2020. [Online]. Available: <https://www.mdpi.com/2076-3417/10/21/7453>

- [18] N. Rafie, A. H. Kashou, and P. A. Noseworthy, "ECG interpretation: Clinical relevance, challenges, and advances," *Hearts*, vol. 2, no. 4, pp. 505–513, Nov. 2021. [Online]. Available: <https://www.mdpi.com/2673-3846/2/4/39>
- [19] M. R. Ginsberg, J. A. Morren, and K. Levin, "Using and interpreting electrodiagnostic tests," *Cleveland Clinic J. Med.*, vol. 87, no. 11, pp. 671–682, Nov. 2020.
- [20] K. B. Kim and H. J. Baek, "Photoplethysmography in wearable devices: A comprehensive review of technological advances, current challenges, and future directions," *Electronics*, vol. 12, no. 13, p. 2923, Jul. 2023. [Online]. Available: <https://www.mdpi.com/2079-9292/12/13/2923>
- [21] A. Laghari, Z. A. Memon, S. Ullah, and I. Hussain, "Cyber physical system for stroke detection," *IEEE Access*, vol. 6, pp. 37444–37453, 2018.
- [22] C. M. Wilkinson, J. I. Burrell, J. W. P. Kuziek, S. Thirunavukkarasu, B. H. Buck, and K. E. Mathewson, "Predicting stroke severity with a 3-min recording from the muse portable EEG system for rapid diagnosis of stroke," *Sci. Rep.*, vol. 10, no. 1, p. 18465, Oct. 2020.
- [23] Y.-A. Choi, S. Park, J.-A. Jun, C. M. B. Ho, C.-S. Pyo, H. Lee, and J. Yu, "Machine-learning-based elderly stroke monitoring system using electroencephalography vital signals," *Appl. Sci.*, vol. 11, no. 4, p. 1761, Feb. 2021.
- [24] M. S. Islam, I. Hussain, M. M. Rahman, S. J. Park, and M. A. Hossain, "Explainable artificial intelligence model for stroke prediction using EEG signal," *Sensors*, vol. 22, no. 24, p. 9859, Dec. 2022.
- [25] K. Rathakrishnan, S.-N. Min, and S. J. Park, "Evaluation of ECG features for the classification of post-stroke survivors with a diagnostic approach," *Appl. Sci.*, vol. 11, no. 1, p. 192, Dec. 2020.
- [26] J. Yu, S. Park, S.-H. Kwon, C. M. B. Ho, C.-S. Pyo, and H. Lee, "AI-based stroke disease prediction system using real-time electromyography signals," *Appl. Sci.*, vol. 10, no. 19, p. 6791, Sep. 2020.
- [27] I. Hussain and R. Jany, "Interpreting stroke-impaired electromyography patterns through explainable artificial intelligence," *Sensors*, vol. 24, no. 5, p. 1392, Feb. 2024.
- [28] J. Yu, S. Park, S.-H. Kwon, K.-H. Cho, and H. Lee, "AI-based stroke disease prediction system using ECG and PPG bio-signals," *IEEE Access*, vol. 10, pp. 43623–43638, 2022.
- [29] N. Goren, J. Avery, T. Dowrick, E. Mackle, A. Witkowska-Wrobel, D. Werring, and D. Holder, "Multi-frequency electrical impedance tomography and neuroimaging data in stroke patients," *Scientific Data*, vol. 5, no. 1, pp. 1–10, Jul. 2018.
- [30] H. Abdi and L. J. Williams, "Principal component analysis," *Wiley Interdiscipl. Rev., Comput. Statist.*, vol. 2, no. 4, pp. 433–459, 2010.
- [31] P. Comon, "Independent component analysis, a new concept?" *Signal Process.*, vol. 36, no. 3, pp. 287–314, Apr. 1994.
- [32] A. Gramfort, "MEG and EEG data analysis with MNE-Python," *Frontiers Neurosci.*, vol. 7, pp. 1–13, 2013.
- [33] J. Machado and A. Balbinot, "Executed movement using EEG signals through a naive Bayes classifier," *Micromachines*, vol. 5, no. 4, pp. 1082–1105, Nov. 2014. [Online]. Available: <https://www.mdpi.com/2072-666X/5/4/1082>
- [34] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017.
- [35] R. E. Schapire, "Explaining AdaBoost," in *Empirical Inference: Festschrift in Honor of Vladimir N. Vapnik*. Cham, Switzerland: Springer, 2013, pp. 37–52.
- [36] T. Chen, T. He, M. Benesty, V. Khotilovich, Y. Tang, H. Cho, K. Chen, R. Mitchell, I. Cano, T. Zhou, M. Li, J. Xie, M. Lin, Y. Geng, Y. Li, and J. Yuan, "XGBoost: Extreme gradient boosting," *R Package*, vol. 1, no. 4, pp. 1–4, 2014.
- [37] S. Zhang, X. Li, M. Zong, X. Zhu, and R. Wang, "Efficient kNN classification with different numbers of nearest neighbors," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 5, pp. 1774–1785, May 2018.
- [38] S. B. Kotsiantis, "Decision trees: A recent overview," *Artif. Intell. Rev.*, vol. 39, no. 4, pp. 261–283, Apr. 2013.
- [39] S. Ratnasingam and J. Muñoz-Lopez, "Distance correlation-based feature selection in random forest," *Entropy*, vol. 25, no. 9, p. 1250, Aug. 2023. [Online]. Available: <https://www.mdpi.com/1099-4300/25/9/1250>
- [40] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas, "Taking the human out of the loop: A review of Bayesian optimization," *Proc. IEEE*, vol. 104, no. 1, pp. 148–175, Jan. 2016.
- [41] A. X. Stewart, A. Nuthmann, and G. Sanguinetti, "Single-trial classification of EEG in a visual object task using ICA and machine learning," *J. Neurosci. Methods*, vol. 228, pp. 1–14, May 2014. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0165027014000752>
- [42] F. Shariaty and S. S. Roy, *MATLAB® for Brain-Computer Interface Systems: Computation and Data Processing*. Boca Raton, FL, USA: CRC Press, 2025.
- [43] J.-N. Lee and K.-C. Kwak, "ECG-based biometrics using a deep network based on independent component analysis," *IEEE Access*, vol. 10, pp. 12913–12926, 2022.
- [44] A. Efendi, Z. Zuraidah, D. S. Susanti, N. N. Debataraja, R. B. E. Wibowo, and S. Handoyo, "Acquiring independent components through hybrid PCA and ICA to enhance the classification performance of decision tree," *Statist., Optim. Inf. Comput.*, vol. 13, no. 5, pp. 1832–1846, Feb. 2025.
- [45] S. Makeig, A. J. Bell, T. Jung, and T. J. Sejnowski, "Independent component analysis of electroencephalographic data," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 8, 1995, pp. 145–151.
- [46] F. Artoni, A. Delorme, and S. Makeig, "Applying dimension reduction to EEG data by principal component analysis reduces the quality of its subsequent independent component decomposition," *NeuroImage*, vol. 175, pp. 176–187, Jul. 2018.

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